

1. Study Design & Cohort definition

Define the research question
study population,
outcome & predictors

2. Exploratory Data Analysis

Explore distributions
Visualize relationships

3. Model selection

Choose algorithms
Complexity–interpretability–performance trade-offs

4. Data Preprocessing & Feature Engineering

Clean data & handle missing data
encode variables, apply transformations
Create features

5. Data splitting

Split the data randomly
(e.g., 75% training, 25% test),
or by time

6. Model Training

Fit the models
Optimize parameters
using cross-validation methods
on the training set
Debug the model if needed

7. Model Evaluation

Evaluate discrimination (ROC–AUC)
calibration (Calibration plots)
validation on the test set

8. Interpretation & Validtion

Analyze the model
Post-hoc explainability methods
for black box models
(SHAP values, partial dependence plots)